Kinetic Theory of Gases

Developed by Daniel Bernoulli (1738) and subsequent progress was made by Joule, Clausius, Maxwell and Boltzmann within 1848-1898.

Assumptions:

(1) The gas is composed of very small discrete particles, called molecules. For a gas, the mass of the molecules are same and different for different gases.

(2) The molecules are moving at random in all directions with variety of speeds. Some are very fast while others are slow.

(3) Due to random motion, the molecules are executing collision with the walls of the container (wall-collision) and also with themselves (intermolecular collisions). These collisions are perfectly elastic, and so there occurs no loss of kinetic energy or momentum of the molecules by these collisions.

(4) The gas molecules are assumed to be point masses, i.e., their size is very small in comparison to the distance they travel.

(5) There exists no intermolecular attractions especially at low Pr., i.e., one molecule can exert Pr. independent of the influence of the other molecules.

(6) The pressure exerted by a gas is due to the incessant and uniform wall-collisions of the molecules. Higher the frequency of the wall-collisions, greater will be the Pr. of the gas. *This explains Boyle's Law since when Vol. is reduced, wall-collisions becomes more frequent and the Pr. is increased.*

(7) Though the molecular speeds are constantly changing due to intermolecular collisions, average kinetic energy ($\bar{\varepsilon}$) of the molecules remains fixed at a given temp. *This explains Charles Law that when temp. is increased, $\bar{\varepsilon}$ is increased, velocities (as $\varepsilon = ½ mc^2$) are increased, wall-collisions becomes more frequent and violent, so Pr. is increased when Vol. is kept constant or Vol. is increased when Pr. is kept constant.*

Idea of root-mean-square speed (c_{RMS}) :-

Let in a gas of N molecules, N_1 have speed c_1, N_2 molecules have speed c_2, N_3 molecules have speed c_3 and so on, then

$$\bar{\varepsilon} = \frac{N_1\varepsilon_1 + N_2\varepsilon_2 + N_3\varepsilon_3 + \cdots}{N}$$

$$= \frac{1}{N}\left[N_1 \times \frac{1}{2} m c_1^2 + N_2 \times \frac{1}{2} m c_2^2 + N_3 \times \frac{1}{2} m c_3^2 + \cdots\right]$$

$$= \frac{1}{2} m \left[\frac{N_1 c_1^2 + N_2 c_2^2 + N_3 c_3^2 + \cdots}{N}\right] = \frac{1}{2} m c_{RMS}^2$$

The term within the bracket is called c_{RMS}^2 (mean-square-speed, $\bar{c}^2$) and root-mean-square speed is defined as

$$c_{RMS} \left(= \sqrt{\bar{c}^2}\right) = \sqrt{\frac{N_1 c_1^2 + N_2 c_2^2 + N_3 c_3^2 + \cdots}{N}}$$

Concept of Temp. and Kinetic Theory:-

The speed (c_{RMS}) is more fundamental than average speed, $\bar{c}$, since it originates spontaneously from the average kinetic energy of the molecules and it depends on the thermodynamic parameter, temp (T) (Postulate 7).

Kinetic theory of gases can provide a concept of temp. (*T*). Average kinetic energy ($\bar{\varepsilon}$) is a measure of temp. Higher the average kinetic energy of a gas, greater will be its temperature. When the molecules are at rest (i.e., $\bar{\varepsilon} = 0$), the temp. will also be 0 K.

Kinetic Theory of Gases

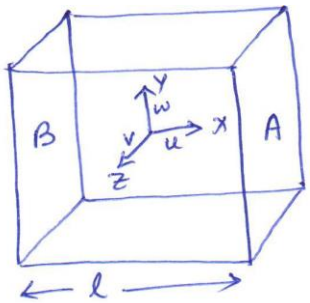
Formulation of Kinetic Equation:- $PV = \frac{1}{3} m N c_{RMS}^2$

The postulates of the kinetic theory enable us to calculate the Pr. of a gas in terms of molecular quantities.

Let us take a cube of edge length *l* containing *N* molecules of a gas of molecular mass *m* and RMS speed is *c* at temp. *T* and Pr. *P*. Let N_1 molecules have speed c_1, N_2 molecules have speed c_2, N_3 molecules have speed c_3, and so on.

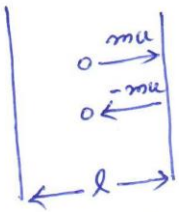
Let us concentrate our attention to a single molecule among N_1 that have resultant speed c_1 and the component velocities are u, v, and w along the X-, Y- and Z-axis, respectively. So,

$$c_1^2 = u^2 + v^2 + w^2$$



Volume of the cube, $V = l^3$

Total surface area $= 6l^2$



change of momentum for the collision

$= mu - (-mu)$

$= 2mu$

Molecule will collide walls A and B with the component velocity *u* and other opposite faces by *v* and *w*.

Change of momentum of the molecule along X-direction for a single collision = mu – (–mu) = 2mu.

The number of collisions suffered by the molecule in unit time between the two opposite walls = u/l

So, the rate of change of momentum for the above type collisions = $2mu \times \frac{u}{l} = \frac{2mu^2}{l}$

Similarly, along y- and z- directions, the rate of change of momenta of the molecule are $\frac{2mv^2}{l}$ and $\frac{2mw^2}{l}$, respectively.

Total rate of change of momentum of the molecule

$$= \frac{2m}{l}(u^2 + v^2 + w^2) = \frac{2mc_1^2}{l}$$

Kinetic Theory of Gases: Formulation of Kinetic Equation

For similar N_1 molecules, the rate of change of momentum $= \frac{2mN_1c_1^2}{l}$

Taking all the molecules of the gas, the total rate of change of momentum

$$= \frac{2mN_1c_1^2}{l} + \frac{2mN_2c_2^2}{l} + \frac{2mN_3c_3^2}{l} + \cdots$$

$$= \frac{2mN}{l}\left(\frac{N_1c_1^2+N_2c_2^2+N_3c_3^2+\cdots}{N}\right) = \frac{2mN\bar{c}^2}{l}$$

According to the Newton's 2nd Law of motion, rate of change momentum due to wall-collision is equal to the force developed on the walls of the gas container. Thus,

$$P \times 6l^2 = \frac{2mN\bar{c}^2}{l}, \quad \text{but } l^3 = V, \text{ volume of the gas in the cube.}$$

P (Pr.) x Area ($6l^2$) = F (force)

So, the kinetic equation of the gas is; $P \times V = \frac{1}{3} m N \bar{c}^2$

Again the equation is $P = \frac{1}{3}\left(\frac{mN}{V}\right)\bar{c}^2$ but $\frac{mN}{V} = d$

d = density of the gas at the given temp. and Pr. So, another form of the kinetic equy. is $P = \frac{1}{3} d \bar{c}^2$. This c is RMS speed of the gas molecules.

These equations are also valid for any shape of the gas container other than cubic one.

Expression of root-mean square speed:-

Let us apply the kinetic equation for 1 mole ideal gas. In that case, mN = mN_A = M, the molar mass of the gas. Ideal gas equation for 1 mole is PV = RT

$$PV = \frac{1}{3} m N C_{RMS}^2$$

Putting in the kinetic equn., we have $RT = \frac{1}{3} M C^2$.

So, $C \left(= \sqrt{\bar{C}^2}\right) = \sqrt{\frac{3RT}{M}}$

This shows that RMS speed depends on the molar mass (M) and temp. (T) of the gas.

Expression of average kinetic energy ($\bar{\varepsilon}$) :-

The kinetic equn. can be written as: $PV = \frac{2}{3} N \left(\frac{1}{2} m \bar{C}^2\right)$

But, $\frac{1}{2} m \bar{c}^2 = \bar{\varepsilon}$, Average kinetic energy of the molecules.

Considering 1 mole ideal gas, $RT = \frac{2}{3} N_A \bar{\varepsilon} \Rightarrow \bar{\varepsilon} = \frac{3}{2}\left(\frac{R}{N_A}\right) T$

Thus, the expression of average kinetic energy of the gas molecules at temp. T is given as : $\bar{\varepsilon} = \frac{3}{2} kT$, where k = Boltzmann const. $= \frac{R}{N_A} = 1.38 \times 10^{-16}$ erg molecules^{-1} K^{-1}

For 1 mole gas, the KE is $\bar{E} = \frac{3}{2} RT$

For n mole of the gas, the KE is $E = \frac{3}{2} nRT$

$\bar{\varepsilon}$ depends on the temp. only and does not depend on the nature of the gas. Thus, most light gas H_2 and very heavy gas UF_6 both have same average KE at a given temp.

Kinetic Theory of Gases: Deduction of the gas laws from kinetic equation

From kinetic eqn., we have $PV = \frac{2}{3} N_A \bar{\varepsilon}$,

but, $\bar{\varepsilon} \propto T$, $\Rightarrow \bar{\varepsilon} = k'T$. Thus, $PV = \left(\frac{2}{3} N_A k'\right) T = \text{const.} \times T$

This equation provides necessary deduction of the gas laws such as Boyle's law, Charles law, etc.
***Deduction of Avogadro's law is shown here*;**

Let us consider two gases under same T, P and V. Thus,

$$PV = \frac{1}{3} m_1 N_1 c_1^2 = \frac{1}{3} m_2 N_2 c_2^2$$

$$\Rightarrow m_1 N_1 c_1^2 = m_2 N_2 c_2^2 \quad \text{— (1)}$$

Again, the two gases have the same average KE at the given T.

So, $\bar{\varepsilon}_1 = \bar{\varepsilon}_2 \Rightarrow \frac{1}{2} m_1 c_1^2 = \frac{1}{2} m_2 c_2^2 \Rightarrow m_1 c_1^2 = m_2 c_2^2$ — (2)

So, dividing the equation (1) by the equation (2), we have $N_1 = N_2$. It means that equal volumes of all gases at the same *T* and *P* contain equal number of molecules and it is Avogadro's law.

Deduction of Graham's law of diffusion (Thomas Graham, Scottish Physical Chemist, 1848):

The rate of diffusion of a gas (r_d) is directly proportional to the speed of the gas.

Thus, $r_d \propto c$, but $c = \sqrt{\frac{3RT}{M}} \Rightarrow r_d \propto \sqrt{\frac{3RT}{M}}$.

This shows that at a given T, $r_d \propto \sqrt{\frac{1}{M}}$.

Thus, at given temp. (*T*), the rate of diffusion is inversely proportional to the square root of molar mass of the gas and this is *Graham's law of diffusion.*

What is the difference between root mean square velocity and average velocity?

The biggest difference is that the root mean square velocity is a **scalar** quantity and the average velocity is a **vector** quantity.

The average velocity is the sum of all of the velocity vectors (added component-wise) divided by the total number of vectors you are adding, i.e.

$$\mathbf{v}_{\text{avg}} = \frac{1}{N}\sum_{i}^{N}\mathbf{v}_i$$

Since this quantity is a vector, it has a direction and length. It also depends on the directions of each individual velocity vector you are considering, since vectors that face opposite directions cancel out.

The root mean square velocity (which I will denote with ν) is the square root of the average length squared of the velocity vectors, i.e.

$$\nu = \sqrt{\frac{1}{N}\sum_{i}^{N}|\mathbf{v}_i|^2}$$

This quantity is just a number, it does not indicate what the average direction of the velocities are, In fact, this quantity is not dependent on the direction of any of the velocities.